# Automated Discovery of an Extremal Black Hole in a 4D Spacetime Simulation

QuantumVerse AI Agent
*QuantumVerse Simulator, Autonomous Discovery Engine*

We report a near-extremal Kerr black hole candidate identified by QuantumVerse's discovery scan. The spin_parameter is 0.999 and the confidence is 50%. Harness test abstract with LaTeX specials: _ % & # $ { } ~ ^ \.

## INTRODUCTION

The QuantumVerse Simulator utilizes a PPO reinforcement learning agent to explore the parameter space of General Relativity. The agent navigates the black hole mass-spin-charge parameter space while respecting the cosmic censorship hypothesis, which forbids naked singularities. Each proposed state is validated by the PhysicsValidator module before acceptance.

## METHODOLOGY

The RL agent explored the parameter space near the extremal limit ($a^2+Q^2 \approx M^2$). The proposed metric was validated against cosmic censorship using the PhysicsValidator module. The final validated parameters were determined to be:

- Mass ($M$): 1.00 $M_\odot$
- Spin ($a$): 0.9990
- Charge ($Q$): 0.0000

The extremality parameter $a^2 + Q^2/M^2$ was computed to verify proximity to the extremal bound without exceeding it.

## RESULTS

The spacetime curvature was evaluated at the probe location using the Kerr metric curvature invariants. The Kretschmann scalar was found to be $K = 1.0000e-06$. The gravitational redshift at the probe radius was calculated as $z = 1.0000e-01$. All curvature invariants remained finite, confirming the absence of a naked singularity.

## CONCLUSION

The AI agent successfully navigated the parameter space to find a stable, near-extremal black hole without violating cosmic censorship. The discovered spacetime satisfies all General Relativity validity checks, with curvature invariants well-behaved outside the event horizon.